

Practical Sheet 10 – Curves and Vector Functions

Cálculo2E

1. Sketch the following curves:
 - (a) $x = 4t - 1, y = t + 2$, in $\mathbb{R}^2$;
 - (b) $x = 2t + 1, y = t^2$, in $\mathbb{R}^2$;
 - (c) $x = \sin t, y = \cos t - 3$, in $\mathbb{R}^2$;
 - (d) $x = 2 \sin t, y = 4 \cos t, z = 1, 0 \leq t \leq 2\pi$, in $\mathbb{R}^3$;
 - (e) $x = \cos t, y = \sin t, z = t/2\pi, -2\pi \leq t \leq 2\pi$, in $\mathbb{R}^3$.
 - (f) For each curve above, compute the velocity vectors for all t .
2. Compute the velocity vectors for all t and the equation of the tangent line at the specified value of t :
 - (a) $(\sin(3t), \cos(3t), 2t^{3/2})$; $t = 1$
 - (b) $(t \sin t, t \cos t, \sqrt{3}t)$; $t = 0$
 - (c) $(2 \cos t, 3 \sin t, t)$; $t = \pi$
 - (d) $(t^2 - 1, \cos t^2, t^4)$; $t = \sqrt{\pi}$
3. Determine an equation of the tangent line to the curves below at the indicated point
 - (a) $x = 1/t, y = \sqrt{t+1}, t_0 = 2$ (in $\mathbb{R}^2$).
 - (b) $x = \theta - \sin \theta, y = 1 - \cos \theta, \theta_0 = \pi/4$ (in $\mathbb{R}^2$).
 - (c) $x = t^2 - 1, y = \cos t^2, z = t^4, t_0 = \sqrt{\pi}$ (in $\mathbb{R}^3$).
4. Consider the curve in $\mathbb{R}^3$ defined by the equations $\frac{x^2}{4} + 2y^2 = 1$ and $z = 2$.
 - (a) Determine a parametric representation of the curve.
 - (b) Determine the tangent line to the curve at the point $P_0 = (\sqrt{2}, \frac{1}{2}, 2)$.
5. For each of the following curves, determine the tangent line at the indicated point.
 - (a) $16x^2 + 9y^2 = 1$ in $\mathbb{R}^2, P_0 = (-\frac{1}{4}, 0)$.
 - (b) $y = 3x - 2$ in $\mathbb{R}^2, P_0 = (\frac{2}{3}, 0)$.
 - (c) $y = \cos(2x)$ in $\mathbb{R}^2, P_0 = (-\frac{\pi}{6}, \frac{1}{2})$.
 - (d) $x = y^3 = z^2 + 1$ in $\mathbb{R}^3, P_0 = (1, 1, 0)$.
6. Consider the surface $S = \{(x, y, z) \in \mathbb{R}^3 : 1 - z = x^2 + 3y^2 \wedge z \geq -2\}$. Let C be the curve in $\mathbb{R}^3$ corresponding to the boundary of the surface S .
 - (a) Represent geometrically the curve C .
 - (b) Determine a parametric representation of the curve C .
7. Suppose a particle follows a path C , described by $(e^t, e^{-t}, \cos t)$ until time $t = 0$, after which it follows the path described by the tangent line to C at that instant. Where is the particle at $t = 2$?

8. Consider the vectorial function $\vec{\sigma}(t) = \frac{t}{2}\vec{i} + \cos(\sqrt{2}t)\vec{j} + \sin(\sqrt{2}t)\vec{k}$. Let C be the curve with parametric equations $x = \frac{t}{2}, y = \cos(\sqrt{2}t)$ e $z = \sin(\sqrt{2}t), t \in [0, 2\pi]$. Assume that curve C is the trajectory of a particle P .
- Determine the unit tangent vector at each point of the curve C .
 - Compute the length of the curve C .
 - Determine the distance traveled by particle P for $t = 2$.
 - Determine the distance traveled by particle P for $t = \frac{\pi}{2}$.
 - At what point is particle P after traveling a distance of $\frac{\pi}{2\sqrt{2}}$.
9. Suppose a particle P moves in space with velocity $\vec{v}(t) = \vec{i} + \cos\left(\frac{1}{\sqrt{2}}t\right)\vec{j} - \sin\left(\frac{1}{\sqrt{2}}t\right)\vec{k}$ and at time $t = 0$ is at position $(1, 0, \sqrt{2})$. Let C be the curve corresponding to the trajectory of particle P .
- Determine the equation of the tangent line to curve C for $t = \frac{\pi}{\sqrt{2}}$.
 - Determine the distance traveled by particle P between $t = 0$ and $t = 3$.